

Project Report: Household Services Application

1. Student Details

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2. Project Overview

Project Title: Enhanced Household Services Application with Vue.js

Problem Statement:

- To develop an improved version of the household services platform with modern frontend technologies. This enhanced application connects customers seeking household services (e.g., plumbing, cleaning, electrical repairs) with qualified professionals. The platform features real-time updates, image uploads for service verification, and email notifications while maintaining a responsive and intuitive user experience.

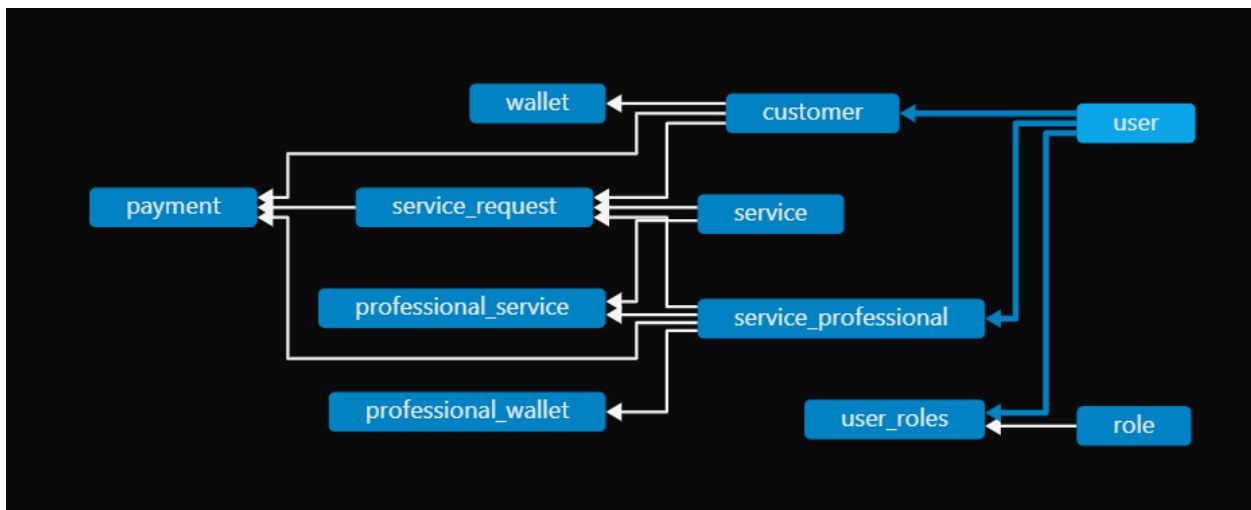
3. Approach

- Implemented a Vue.js frontend for improved reactivity and component-based architecture
- Developed responsive UI using Bootstrap and custom CSS for cross-device compatibility
- Integrated Cloudinary for service-related image uploads and storage (before/after photos)
- Implemented email notification system using Mailhog for service status updates
- Enhanced user authentication with session management and secure password handling
- Created real-time service tracking with status updates
- Developed advanced search and filtering options (by service type, rating, location, availability)
- Implemented professional verification system with document upload capabilities
- Added customer review and rating system with image attachments

4. Frameworks and Libraries Used

- **Vue.js (CDN):** Frontend JavaScript framework for building interactive interfaces
- **Bootstrap:** CSS framework for responsive design and UI components
- **Custom CSS:** Additional styling for unique application requirements
- **Mailhog:** Email testing tool for development environment to handle notification system
- **Cloudinary:** Cloud-based image management solution for storing service-related photos
- **Flask:** Backend framework for building web applications
- **SQLAlchemy:** ORM (Object-Relational Mapping) tool for database interactions
- **SQLite:** Database management system for storing application data
- **Jinja2:** Template engine for rendering dynamic HTML content
- **Flask-RESTful:** Extension for Flask that adds support for quickly building REST APIs

5. ER Diagram



6. API Resource Endpoints

Admin Endpoints

- **URL:** `/api/service_requests/stats`
 - **Method:** GET
 - **Description:** Retrieves an overview of service requests.
- **URL:** `/api/service_requests/monthly`
 - **Method:** GET
 - **Description:** Retrieves monthly trends of service requests.
- **URL:** `/api/services/popular`
 - **Method:** GET
 - **Description:** Retrieves the top services based on the number of bookings.
- **URL:** `/api/professionals/ratings`
 - **Method:** GET

- **Description:** Retrieves the ratings of professionals.
- **URL:** `/api/service_requests/pincode_distribution`
 - **Method:** GET
 - **Description:** Retrieves the distribution of service requests by pincode.
- **URL:** `/api/admin/generate_report`
 - **Method:** GET
 - **Description:** Generates and downloads the CSV or Excel report for the admin.

Customer Endpoints

- **URL:** `/api/customers`
 - **Method:** GET
 - **Description:** Retrieves a list of all customers.
- **URL:** `/api/customers/int:customer_id`
 - **Method:** GET
 - **Description:** Retrieves details of a specific customer.
- **URL:** `/api/customer/profile/int:customer_id`
 - **Method:** GET,PUT
 - **Description:** Retrieves the profile of a specific customer,Updates the profile of a specific customer.
- **URL:** `/api/customer-service-summary/int:user_id`
 - **Method:** GET
 - **Description:** Retrieves the service summary for a specific customer.
- **URL:** `/api/customer/payments`
 - **Method:** GET
 - **Description:** Retrieves the payment history and wallet balance for the customer.
- **URL:** `/api/customer/generate_report`
 - **Method:** GET
 - **Description:** Generates and downloads the CSV or Excel report for the customer.

Professional Endpoints

- **URL:** `/api/service_professionals`
 - **Method:** GET
 - **Description:** Retrieves a list of all service professionals.
- **URL:** `/api/service_professionals/int:professional_id`
 - **Method:** GET
 - **Description:** Retrieves details of a specific service professional.
- **URL:** `/api/professional/profile/int:professional_id`
 - **Method:** GET,PUT
 - **Description:** Retrieves the profile of a specific service professional, Updates the profile of a specific service professional.

- **URL:** `/api/professional/profile/int:professional_id/services`
 - **Method:** GET,PUT
 - **Description:** Retrieves the services offered by a specific service professional,Updates the services offered by a specific service professional.
- **URL:** `/api/professional-service-summary/int:professional_id`
 - **Method:** GET
 - **Description:** Retrieves the service summary for a specific service professional.
- **URL:** `/api/professional/payments`
 - **Method:** GET
 - **Description:** Retrieves the payment history and wallet balance for the professional.
- **URL:** `/api/professional/generate_report`
 - **Method:** GET
 - **Description:** Generates and downloads the CSV or Excel report for the professional.

7. Presentation Video

Drive Link:

https://drive.google.com/drive/u/0/my-drive?q=type:video%20parent:0ANOFXNuguJ_LUk9PVA

8. Conclusion

The enhanced household services application successfully builds upon the foundation of the initial project by implementing modern frontend technologies and additional features. The Vue.js integration provides a more responsive and dynamic user experience, while Cloudinary support enables visual documentation of services. The email notification system powered by Mailhog keeps all stakeholders informed about service status changes.

The application demonstrates significant improvements in user experience, functionality, and technical implementation. Future enhancements could include a mobile application, integration with payment gateways, and a chatbot for instant customer support.